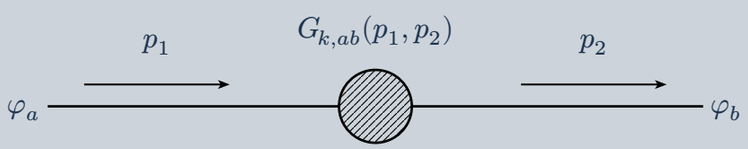
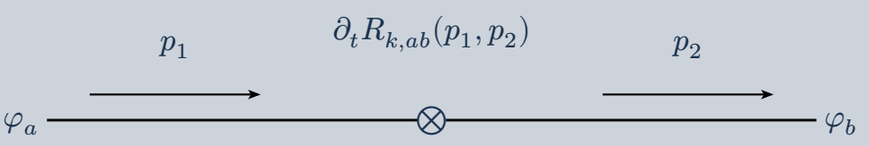
Read a functional renormalization-group diagram one symbol at a time. Lines connect fields; vertices encode their interactions.

1 Propagator



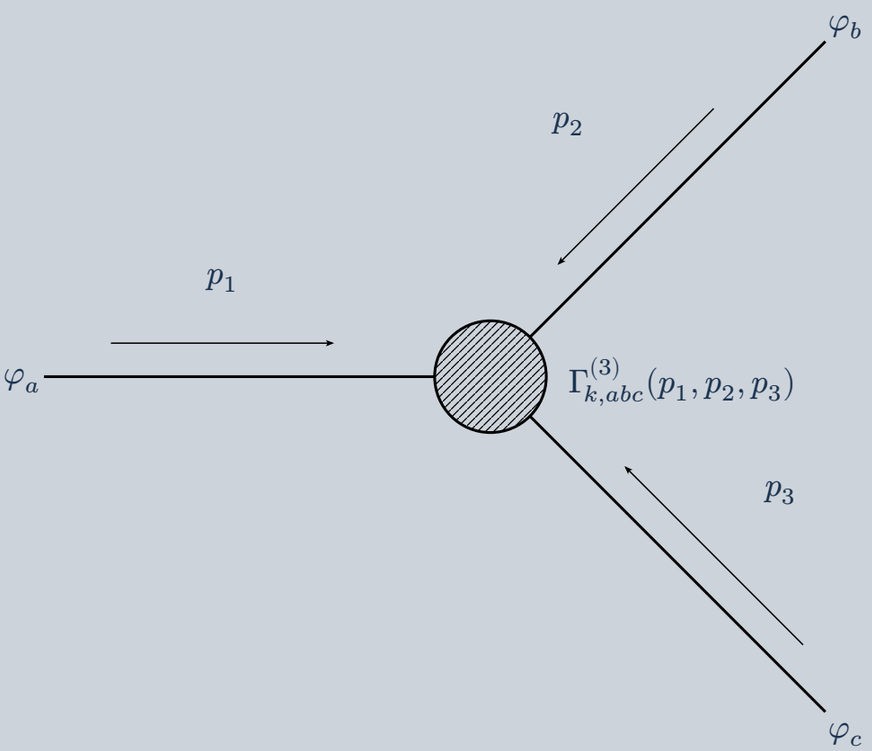
The full propagator G_k describes a two-point correlation at scale k. "Full" includes interaction corrections.

2 Regulator insertion



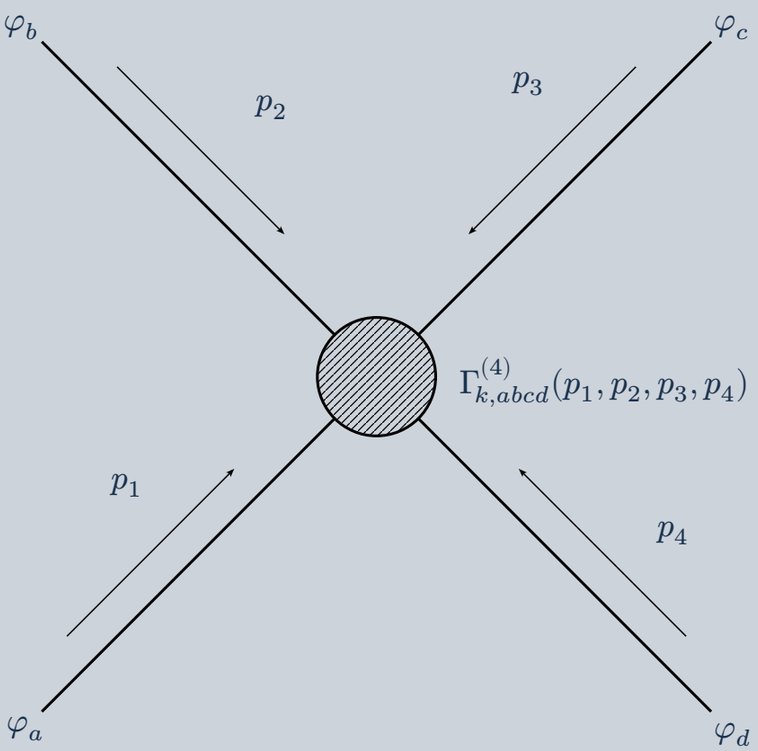
The circled cross means $\partial_t R_k$ with RG time $t = \ln(k/\Lambda)$: the change of the momentum cutoff. It is an insertion on a line, not a new particle.

3 Three-point vertex



$\Gamma_k^{(3)}$ couples three field legs. Each p_i labels a momentum; letter indices label field components.

4 Four-point vertex



$\Gamma_k^{(4)}$ couples four field legs. The hatched disc marks a dressed vertex, including fluctuation effects.

Read the connections, then the labels. k is a coarse-graining scale; Γ_k is the effective action. Repeated internal indices are summed and loop momenta are integrated.